August 7, 2026

Dear Hiring Team,

I am writing to apply for the Data Scientist I position at Miovision. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, SQL-backed systems, statistical model evaluation, machine learning, geospatial data processing, and visualization, I am excited by the opportunity to help Miovision turn real-world transportation data into useful, reliable insights for safer and smarter communities.

My graduate research focused on building data-driven software for complex Earth observation datasets. Working with BigGeo, Mitacs, and the University of Calgary, I developed Python and C++ pipelines for data ingestion, preprocessing, storage, retrieval, and analysis. I automated SAR and optical imagery preprocessing with GDAL, Rasterio, SNAP, and external data APIs, and built DGGS-based workflows that transformed satellite imagery into structured tensor files for downstream analysis. This gave me practical experience with messy real-world data, data quality, reproducible workflows, and the details that determine whether an analysis is actually trustworthy.

I also bring direct machine learning and model evaluation experience. In my DEM super-resolution project, I trained PyTorch ResNet models, collected and processed terrain data using Google Earth Engine and Rasterio, and evaluated results with statistical and terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. I also built a Streamlit-based workflow for data collection and preprocessing and a Polyscope-based 3D visualization tool to inspect model outputs. That combination of modeling, evaluation, visualization, and communication aligns closely with Miovision's need for someone who can explore large datasets, identify patterns, improve data quality, and explain findings clearly.

My backend experience adds another useful layer for a customer-facing analytics platform. As a back-end developer, I built Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL, implemented authentication flows, and worked with structured service and payment data. As a teaching assistant at the University of Calgary, I supported students in big data applications, data-at-scale workflows, SQL, and database design, which strengthened my ability to communicate technical ideas to people with different backgrounds.

I am particularly interested in Miovision because the work is applied and measurable: transportation data, infrastructure decisions, customer-facing models, and products that can improve how cities move. I have not worked directly with Snowflake, Sigma, or Dash yet, and my AWS exposure has been through coursework and projects rather than enterprise production ownership, but I bring a strong base in Python, SQL, machine learning, geospatial analysis, data pipelines, Git, and careful technical communication. I learn new tools quickly and enjoy digging into a dataset until I understand both the signal and the limitations.

Thank you for considering my application. I would welcome the opportunity to discuss how my data science, machine learning, visualization, and real-world geospatial data experience can contribute to Miovision's Software Engineering team.

Sincerely,
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